

# Phase 2Requirement Analysis

## Requirements Specification Document

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|--------|---------------|
| Role   | Name          |
| Member | Boora Lakshmi |

### Software Dependencies

- Operating System: Windows 10/11, macOS, or Ubuntu Linux.
- Runtime Environment: Python 3.12 or higher.
- External Packages: Flask, pandas, numpy, scikit-learn, xgboost, seaborn, matplotlib (managed via requirements.txt).

### Hardware Configurations

- Processor: Intel i5 / AMD Ryzen 5 or higher (Multi-core preferred for faster hyperparameter tuning).
- RAM: Minimum 8 GB (16 GB recommended during the RandomizedSearchCV phase).
- Storage: Minimum 2 GB of free disk space for datasets, virtual environments, and dependencies.

### Data Specifications

The system utilizes the loan prediction.csv dataset.

- Total Records: 614 entries.
- Total Columns: 13 features.
- Datatypes: 12 features are strictly encoded as int64 (e.g., Gender, Married, Education, ApplicantIncome). 1 feature is str (Loan ID, which is dropped prior to model training).
- Target Variable: Loan Status (Integer encoded: 1 for Approved, 0 for Rejected).